

Data and Code Documentation for “Assessing the Heterogeneous Impact of Economy-Wide Shocks: A Causal Machine Learning Approach Applied to Colombian Firms”

Overview

This document provides an overview of the data and code files that accompany the paper “*Assessing the Heterogeneous Impact of Economy-Wide Shocks: A Causal Machine Learning Approach Applied to Colombian Firms*”. It explains the software requirements, instructions for running the code to reproduce all tables and figures (including any appendix results), the organization of the code and data files, descriptions of datasets used (with notes on access), and other relevant information for replication.

The online replication package includes the final analysis datasets `final_data_18.RData`, `final_data_19.RData`, and `final_data_20.RData`. These are the starting point for reproducing the main tables and figures described from Section 4, subsection “Reproduction of Tables and Figures”, onward. The raw Colombian customs microdata are not redistributed in this package. However, the package includes the code needed to construct the included final analysis datasets from the raw data that can be downloaded from the sources described in Section 5.2.

1 Software Requirements and Computational Environment

- **Operating System:** The code was tested on *Windows 11 Pro (64-bit)* and *macOS 26.1 (25B78)*. The replication scripts should run on any OS with the required software.
- **Stata:** Stata/SE 19.5 for Windows and StataNow/SE 19.5 for Mac (Apple Silicon).
- **R/RStudio:** R 4.5.3 was used for the Windows checks. The Mac environment used RStudio 2025.09.2+418.

2 Replication Instructions

To replicate the results of the paper, follow these steps:

1. **Setup:** Ensure that all software listed in the previous section is installed. Download the replication package and locate the main directory. The replication package folder is called `replication_code_online`.
2. **Data Placement:** The package includes the final analysis datasets in `data/data_out/`. To reproduce the main paper results from Section 4.3 onward, start from `final_data_18.RData`, `final_data_19.RData`, and `final_data_20.RData`. To rebuild these files from the raw data, download the raw export and import microdata from the sources listed in Section 5.2, place them in the structure expected by the data-construction scripts, and run the code described in subsections 4.1 and 4.2.
3. **Configure Paths (if needed):** If the code requires specifying file paths or other configuration, adjust those in the configuration script
4. **Output:** After successful execution, all tables and figures will be generated and saved in the designated folder (and/or displayed in the console as applicable). Each folder (and file) is clearly named (e.g., `Table1_results.txt`, `Figure2_plot.pdf`) for easy reference.

3 File Structure and Organization

The replication package is organized as follows (main directories and files). Figure files are contained in the following directories:

- `README.pdf` – This document, describing the data and code for replication.
- `replication_code_online/` – Directory containing all code (scripts and programs) used in the analysis.
 - [Fig_1](#) – Contains the file `individual_TE.R` which is the one for reproducing the individual Treatment Effects and Fig. 1 in the main text. The CI can be recovered by running the code in the folder `bootstrap_for_CI`
 - [Fig_2](#) – Contains the file `heterogeneity_quarterly_industry.R`, which reproduces the quarterly average COVID-19 effect by industry and Fig. 2 in the main text. The code uses the original dataset to compute the point estimates and the bootstrap dataset to recover the standard errors and confidence intervals (both can be found in the code reproducing [Tab_4](#)).
 - [Fig_3](#) – Contains the file `fig_3_annual_spe.R` which is the one for reproducing the individual Treatment Effects and Fig. 3 in the main text.
 - [Fig_4_and_7](#) – Contains two sub-directories namely `SAM_minus_SUM` and `Y_minus_SUM` representing respectively the construction of the monthly SPE figure for SAM-SUM ([Fig_4](#)) and the monthly SPE figure for Y-SUM ([Fig_7](#)).
 - [Fig_5](#) – Contains the code to reproduce the comparison between the SAM-SUM and Y-SUM average treatment effects by month ([Fig_5](#)).

- [Fig_6](#) – Contains the code to reproduce the percentile-by-quarter comparison of the estimated effects used in [Fig_6](#).
- [Fig_8](#) – Contains the code to reproduce the heterogeneous treatment effects estimated in the A sample and predicted in the M sample for the month of January ([Fig_8](#)). Notice that this partly reproduces the code in `blp_clan_gates_month1` inside `section_4_point_5`.

Table files are contained in the following directories ([Tab_1](#) and [Tab_2](#) are explicative only and did not result from the use of data.):

- [Tab_3](#) – Contains the necessary files for reproducing the Goodness of Fit (GoF) [Tab_3](#) in the main paper.
- [Tab_4](#) – Contains the replication files for the differences in means of the estimated treatment effect and other covariates between the group of more affected and the group of less affected firms (CaDiff) in the case of SAM-SUM.
- [Tab_5](#) – Following the same structure of the code in [Tab_4](#), it contains the replication files for the differences in means of the estimated treatment effect and other covariates between the group of more affected and the group of less affected firms (CaDiff) in the case of Y-SUM.
- [Tab_6](#) – Contains the BLP for each month and for each learner. Namely the code runs the Best Linear Predictor (BLP) regression on the main sample M:

$$\tilde{Y}^{ATE} = \beta_1 + \beta_2 \cdot (\hat{\Delta}(X) - E[\hat{\Delta}(X)]) + u$$

where: (i) $\tilde{Y}^{ATE}$ is the AIPW pseudo-outcome; (ii) $\hat{\Delta}(X)$ is the learner-specific predicted CATE estimated in the auxiliary sample A and evaluated in the main sample M; β_2 is the coefficient of interest represented in the Table and constructed via this code.

- [Tab_7](#) – Contains the replication code for [Tab_7](#), i.e. it constructs median GATES coefficients with their estimated median p-value across 100 A-M splits for the GATEs considering different estimators of CATE.
- [Tab_8](#) – With a similar structure of [Tab_4](#) and [Tab_5](#) it differs from the latter in that it contains the CaDiff for different types of estimators (in separate files for T-Learner, S-Learner, R-Learner, DR-Learner, Causal Forests).

Finally, `section_4_point_5` contains the auxiliary code necessary to reproduce the Section 4.5 monthly heterogeneity exercises, including [Fig_9](#), [Figure Appx.1](#), [Figure Appx.2](#), and [Figure Appx.3](#). Related Section 4.5 tables and figures with dedicated folders are documented separately in [Fig_8](#), [Tab_6](#), [Tab_7](#), and [Tab_8](#).

- `data/` – Directory for datasets.

4 Mapping of Code to Results

4.1 From original dta to inputs of data_in/

The online package does not redistribute the original transaction-level customs files. It does, however, include the code needed to construct the final analysis datasets from the raw data that can be downloaded using the links provided in Section 5.2. The final analysis datasets used by the main replication scripts, `final_data_18.RData`, `final_data_19.RData`, and `final_data_20.RData`, are included in `data/data_out/`. Changes with respect to the version of the raw datasets used in the present work are expected as the Colombian agency of statistics periodically uploads new data and may update old ones.

We use monthly export transaction data from the Colombian Customs Office (DIAN) for 2018–2020, originally provided as raw `/X/Expo_YEAR/month.dta` files. Each file contains transaction-level information, including exporter identifiers, dates, 10-digit HS product codes, origin departments, transport modes, destination countries, and FOB export values in U.S. dollars.

The do-files process the raw monthly `/X/Expo_YEAR/month.dta` files by importing and cleaning each dataset, standardizing variable names and formats, and harmonizing country codes and firm identifiers across months. Cleaned monthly files are then appended within each year to construct annual datasets. The resulting files, `months_2018.dta`, `months_2019.dta`, and `months_2020.dta`, are saved in the `data_in` folder and constitute the input data used in the empirical analysis.

4.2 Construction of the dataset

Refer to the folder `data/`. The latter is structured as follows: (i) `clean_data_final.R` and `f_locals.R` are the R files that take as input the files in `data_in/` to construct the datasets outputted in `data_out/`; (ii) the folder `data_in/` contains the input data used as input of `clean_data_final.R`; the folder `data_out/` contains the resulting datasets used through the main analyses. The main replication exercises described below start from the included files in `data_out/`. In what follows we detail the content of each of the mentioned folders.

- `data_in/` contains the following:
 - A folder called `from_original_to_months` – Contains the following:
 - * `Expo_year` folders (where year take values 2017-2020). The latter are export transactions by Colombian companies for 2014–2020, reported by the Colombian National Administrative Department of Statistics (DANE). (Source)
 - * `Impo_year` folders (where year take values 2017-2020). The latter are import transactions by Colombian companies for 2014–2020, reported by the Colombian National Administrative Department of Statistics (DANE). (Source)
 - * `do_months.v0.do` and `do_months_imports.do` are used for creating `months_2018.dta`, `months_2019.dta`, `months_2020.dta` starting from the original csv files.

- `months_2018.dta`, `months_2019.dta`, `months_2020.dta` – The cleaned monthly datasets adopted to construct the final datasets employed in the analyses. Used in: `clean_data_final.R`.
- `continents_dictionary.csv` – Authors created data that generates a dictionary between countries and continents.
Used in: `clean_data_final.R`.
- `continent_iso.csv` – Country to continent dataset, created by authors.
- `economic_support_index.csv` – Contains monthly information from The Oxford Covid-19 Government Response Tracker (OxCGRT) on year 2020 from Hale et al. (2020). This index summarizes the economic policies response to COVID-19 by countries in the world.
Used in: `clean_data_final.R`.
- `government_response_index.csv` – Contains monthly information from The Oxford Covid-19 Government Response Tracker (OxCGRT) on year 2020 from Hale et al. (2020). This index summarizes the strictness of ‘lockdowns’ to COVID-19 imposed by countries in the world.
Used in: `clean_data_final.R`.
- `containment_health_index.csv` – Contains monthly information from The Oxford Covid-19 Government Response Tracker (OxCGRT) on year 2020 from Hale et al. (2020). This index summarizes health policies to COVID-19 by countries in the world.
Used in: `clean_data_final.R`.
- `stringency_index.csv` – Contains monthly information from The Oxford Covid-19 Government Response Tracker (OxCGRT) on year 2020 from Hale et al. (2020). This index summarizes the overall government response to COVID-19 by countries in the world.
Used in: `clean_data_final.R`.
- `industry_dictionary.RData` – Processed data that generates a dictionary between sectors and industries. Generated in `sector_dictionary_reduced_names.R`
Used in: `clean_data_final.R`.
- `data_out` – Contains the resulting datasets (RData) adopted in the main analyses, called `final_data_18`, `final_data_19`, `final_data_20`.

4.3 Reproduction of Tables and Figures

In what follows we will briefly describe how to reproduce the Tables and Figures contained in the paper’s main text. We will do this in paragraphs to avoid the presence of too many subsections. We will follow the order of appearance in the main text of the paper.

Reproducing Table 3 Tab_3 reproduces Table 3 in the main paper, “Goodness of Fit for SUM and SAM in 2018/19 and 2019/20”. The three panels of the paper are generated by the following folders, in the order in which they appear in the table:

- 2018_2019_SUM.GoF: reproduces the first panel, “Goodness of Fit for SUM in 2018/19”. Run CV_18_19_SUM.R to generate lasso_preds_18.RData, ridge_preds_18.RData, rf_preds_18.RData, and logit_preds_18.RData; then run OBES.GoF_18_19_Table3.R.
- 2019_2020_SUM.GoF: reproduces the second panel, “Goodness of Fit for SUM in 2019/20”. Run CV_18.R to create the 2018 tuning parameters, then SUM_predictions_19_20_SUM.R to generate SUM_preds_lasso.RData, SUM_preds_ridge.RData, SUM_preds_rf.RData, and SUM_preds_logit.RData; then run OBES.GoF_19_20_SUM_Table3.R.
- 2019_2020_SAM.GoF: reproduces the third panel, “Goodness of Fit for SAM in 2019/20”. Run CV_19_SAM.R to generate lasso_preds_19.RData, ridge_preds_19.RData, rf_preds_19.RData, and logit_preds_19.RData; then run OBES.GoF_19_20_SAM_Table3.R.

The prediction scripts use the included data/data_out/final_data_18.RData and data/data_out/final_data_19.RData. The goodness-of-fit scripts compute RMSE and AUC and save CSV files named table3_gof_18_19_sum.csv, table3_gof_19_20_sum.csv, and table3_gof_19_20_sam.csv in their respective folders.

Reproducing Fig.1 Fig.1 reproduces Figure 1 of the main text. The folder combines the prediction files generated for the SUM and SAM exercises and uses them to estimate the individual treatment effect of COVID as the difference between the shock-aware prediction and the shock-unaware prediction for the same firm and month. In the code, this treatment effect is computed as `pred.SAM - pred.SUM`. The figure reports the monthly average of this difference for Logit-LASSO and Random Forest, together with bootstrap confidence intervals.

Fig_1/

```
|-- individual_TE.R
|-- f_locals.R
'-- bootstrap_for_CI/
    |-- bootstrap_lasso_SUM_SAM.R
    '-- bootstrap_rf_SUM_SAM.R
```

`individual_TE.R` is the main script for the figure. It loads the prediction files created when reproducing Table 3. In particular, it uses the SUM prediction files from Tab_3/2019_2020_SUM.GoF and the SAM prediction files from Tab_3/2019_2020_SAM.GoF. The script renames the relevant prediction variables, merges the two sets of predictions by `id`, `month`, and `export_future`, and computes the individual treatment effect for each observation. It then averages this treatment effect by month and plots the resulting monthly series. Figure 1 is produced for LASSO and Random Forest.

The prediction input files are:

- SUM_preds_lasso.RData

- `SUM_preds_rf.RData`
- `lasso_preds_19.RData`
- `rf_preds_19.RData`

The script writes `intervals_lasso_rf_month_r3.pdf`, `intervals_lasso_rf_month_r3.png`, and `fig1_values.csv` in the `Fig_1` folder.

The folder `bootstrap_for_CI` contains the code used to obtain bootstrap confidence intervals for the treatment-effect estimates shown in Figure 1. The online package does not include the bootstrap output files, because they can be large and time-consuming to generate. They are reproduced by running `bootstrap_lasso_SUM_SAM.R` and `bootstrap_rf_SUM_SAM.R`; these scripts save `lasso.pred_SUM.boot.RData`, `lasso.pred_SAM.boot.RData`, `rf.pred_SUM.boot.RData`, and `rf.pred_SAM.boot.RData` in `Fig_1/bootstrap_for_CI`. Once these files exist, `individual_TE.R` uses them to add the confidence intervals to Figure 1.

More specifically, the bootstrap code reproduces the two prediction exercises already described for Table 3, but inside repeated resamples. For `SUM`, the model is trained using the pre-COVID specification and tuning choices; for `SAM`, the model is estimated in a shock-aware way using the 2019 sample and cross-validation within that sample. The bootstrap output therefore provides repeated draws of `pred_SUM` and `pred_SAM`, from which confidence intervals for the monthly treatment effects can be constructed.

Reproducing Fig_2 Fig_2 reproduces Figure 2 of the main text, which reports the quarterly average COVID-19 effect by industry using the Logit-LASSO predictions. The treatment effect is computed at the observation level as the difference between the shock-aware and shock-unaware predicted probabilities, i.e. `pred_SAM - pred_SUM`. The figure then averages this difference by quarter and industry and reports bootstrap confidence intervals.

Fig_2/

```
'-- heterogeneity_quarterly_industry.R
```

`heterogeneity_quarterly_industry.R` is the main script for the figure. It uses the heterogeneity datasets generated in the first part of the Table 4 replication code: `data_heterogeneity_original.csv` for the original estimates and `data_heterogeneity_boots.csv` for the bootstrap replications. If the corresponding `.dta` files are used instead, the script reads the same variables from those files. These large generated files are intermediate outputs of the Table 4 workflow rather than raw data files. The reader is invited to consult paragraph **Reproducing Table 4** for the details of the construction of these datasets.

The script loads the prediction variables `pred_SAM` and `pred_SUM`, the month, the industry classification, and the bootstrap identifier for the bootstrap file. It constructs quarters from the monthly variable, computes `pred_SAM - pred_SUM`, and keeps the industry categories reported in the paper figure. It then computes the original quarterly industry means and uses the bootstrap distribution of the same means to estimate standard errors and 95% confidence intervals.

The main inputs are therefore:

- `data_heterogeneity_original.csv`
- `data_heterogeneity_boots.csv`

The script writes `intervals_lasso_industry_quarter_r_NEW.png`, `intervals_lasso_industry_quarter_r_NEW.pdf`, and `fig2_industry_quarter_values.csv` in the Fig_2 folder.

Reproducing Fig_3 Fig_3 reproduces Figure 3 of the main text, which reports the annual Sorted Partial Effects (SPE) of COVID-19 on firms’ export probability together with the Average Partial Effect (APE) and their bootstrap confidence intervals. The treatment effect is defined at the firm level as the difference between the shock-aware and shock-unaware predicted probabilities, i.e. `pred_SAM - pred_SUM`.

Fig_3/

`-- fig_3_annual_spe.R`

`fig_3_annual_spe.R` is the main script for the figure. It loads two input files, and namely `data_heterogeneity_original.csv` and `data_heterogeneity_boots.csv`. The first file contains the original sample used to compute the firm-level treatment effect and its distribution. The second file contains the bootstrap replications used to estimate standard errors and confidence intervals. These large generated files are intermediate outputs of the Table 4 workflow rather than raw data files. The reader is invited to consult paragraph **Reproducing Table 4** for the details of the construction of the latter. Also, the user should run the first part of **Reproducing Table 4** in order to be able to construct Fig_3.

The script first computes the treatment effect for each observation as `pred_SAM - pred_SUM`. It then constructs the SPE curve by taking the empirical quantiles of this treatment effect over the percentile grid from 1 to 100. The APE is computed as the mean treatment effect in the original sample. Next, the script uses the bootstrap sample to compute the standard error of the SPE at each percentile and the standard error of the APE. These are then used to construct 95% confidence intervals. Finally, the script plots the SPE curve, its confidence band, and the APE with its confidence interval as horizontal reference lines.

The main inputs are therefore:

- `data_heterogeneity_original.csv`
- `data_heterogeneity_boots.csv`

The script writes `SPE_fig3_lightblue.pdf`, `SPE_fig3_lightblue.png`, `fig_3_SPE.pdf`, and `fig3_spe_values.csv` in the Fig_3 folder. The figure shows:

- the red SPE curve across percentiles,
- the shaded bootstrap confidence band around the SPE,
- the black horizontal APE line,
- and the dashed confidence bounds for the APE.

Reproducing Fig.4: SAM_minus_SUM. The latter code is localized in Fig.4.and.7 in the subfolder called `SAM_minus_SUM`. This code reproduces Fig.4. In itself the folder contains other sub-folders that should be run in the following order: (1) `SUM_mar26`, (2) `SAM_mar26`, (3) `bootstraps_mar26` and finally (4) `SPE_mar26` plots Figure 4.

- `SUM_mar26`.

The file `CV_18.R` performs the 2018 cross-validation step used for the shock-unaware prediction exercise. It loads `final_data_18.RData`, cleans the data, and runs month-by-month 5-fold cross-validation for LASSO and Ridge in order to select the optimal penalty parameters. Its output files are `lasso.lambda.months.18.RData` and `ridge.lambda.months.18.RData`.

`SUM_predictions_19_20_SUM.R` uses the monthly tuning parameters selected in `CV_18.R` to generate shock-unaware predictions. It loads `final_data_18.RData`, `final_data_19.RData`, and `lasso.lambda.months.18.RData`, harmonizes the set of covariates across the two years, estimates the monthly LASSO model on the 2018 sample, and predicts export probabilities for the 2019 sample. Its output file is `SUM_preds_lasso.RData`.

- `SAM_mar26`.

`CV_and_predictions_19_SAM.R` combines two steps in a single script. First, it loads `final_data_19.RData`, cleans the data, removes constant variables, creates interaction terms for the logit specification, and then runs month-by-month 5-fold cross-validation to select the optimal LASSO penalty for each month. The output of this first step is `lasso.lambda.months.19.RData`. In the second step, the script reloads the 2019 data, applies the same preprocessing, imports `lasso.lambda.months.19.RData`, and then performs month-by-month 5-fold cross-validated prediction within the 2019 sample. The resulting out-of-sample predicted probabilities and predicted classes are saved in `lasso_preds_19.RData`. Thus, the main input file is `final_data_19.RData`, and the main output files are `lasso.lambda.months.19.RData` and `lasso_preds_19.RData`.

`SAM_predictions_removeconst_outsideloop.R` performs the same overall SAM prediction exercise but in a more efficient way. It again loads `final_data_19.RData`, preprocesses the data, imports the previously estimated monthly LASSO penalties from `lasso.lambda.months.19.RData`, and generates month-by-month 5-fold cross-validated predictions. The main difference is computational: instead of removing constant variables and rebuilding the model matrix separately inside each fold, it does this once at the month level before entering the fold loop. This avoids repetition while leaving the prediction logic unchanged. Its main input files are `final_data_19.RData` and `lasso.lambda.months.19.RData`, and its output file is `lasso_preds_19.RData`.

- `bootstraps_mar26`.

`bootstrap_wlambdacv_and_subsampling_mar26.R` generates the bootstrap prediction files used to estimate uncertainty around the treatment effects. It loads `final_data_18.RData` and `final_data_19.RData`, prepares comparable covariates for the two years, and then repeats the `SUM` and `SAM` prediction exercises month by month

inside a bootstrap loop. In each bootstrap replication, the code draws a 50% subsample with replacement from the relevant training data, re-estimates the LASSO penalty parameter by 5-fold cross-validation using `cv.glmnet`, and then predicts export probabilities. For **SUM**, the model is trained on bootstrapped 2018 data and used to predict the 2019 observations. For **SAM**, the model is trained on a bootstrapped 2019 sample and used to predict the fixed 2019 month-specific sample. The script stores these bootstrap predictions separately for the shock-unaware and shock-aware models.

The main input files are `final_data_18.RData`, `final_data_19.RData`, and `f_locals.R`. The main output files are `lasso.pred_SUM.boot.subsamp.cvlam_marzo26.RData` and `lasso.pred_SAM.boot.subsamp.cvlam_marzo26.RData`, which contain the bootstrap prediction draws for the **SUM** and **SAM** models, respectively.

- **SPE_mar26.**

The file `SPE.fig.R` is the final script that plots Figure 4. It loads the original **SUM** and **SAM** LASSO prediction files, computes the monthly individual treatment effect as `pred_SAM - pred_SUM`, and constructs the monthly SPE curves over the percentile grid from 0.02 to 0.98. It then loads the bootstrap prediction files from `bootstraps_mar26`, aligns the **SUM** and **SAM** bootstrap predictions by firm, month, and bootstrap replication, and uses them to compute the bootstrap confidence bands for the SPE and APE. The script writes `fig_4_goodlegend.pdf`, `spe_sum.sam.fig4.pdf`, and `fig4.spe.values.csv` in the `SAM_minus_SUM` folder.

Reproducing Table 4 For reproducing Table 4 the first step is to construct the dataset `data_heterogeneity_original.csv`. The first step to do so is to construct the dataset `data_heterogeneity.RData`.

The latter is done in the folder `construction_data_heter/`.

Namely the file `data_heterogeneity_mar26.R` inside `construction_data_heter/`, constructs the dataset used for the heterogeneity analysis by combining the original 2019 firm-level characteristics with the LASSO predictions from the **SUM** and **SAM** models. The key intuition is that, for each exporter and month, the script puts side by side the shock-unaware prediction (**SUM**) and the shock-aware prediction (**SAM**), so that heterogeneity in treatment effects can later be studied across firms with different characteristics.

More specifically, the script loads the original LASSO prediction files `SUM_preds_lasso.RData` and `lasso_preds_19.RData` (constructed when reproducing Fig.4, see above), together with the 2019 firm-level dataset `final_data_19.RData` and the auxiliary file `info_mode.RData`. It preprocesses the 2019 data, removes constant variables, and creates the interaction terms used in the logit specification. It then merges the **SUM** and **SAM** predictions by firm and month, reshapes them into a long format, and creates an indicator `covid_aware` equal to one for the **SAM** prediction and zero for the **SUM** prediction. After that, it merges these prediction records with the corresponding exporter characteristics from `final_data_19` and with additional modal information from `info_mode`. Missing character and factor values are recoded as `missing` so that the resulting dataset is ready for later regression and sorting steps in the heterogeneity analysis.

The main input files are `SUM_preds_lasso.RData`, `lasso_preds_19.RData`, `final_data_19.RData`, and `info_mode.RData`. The output file is `data_heterogeneity.RData`.

Once `data_heterogeneity.RData` is constructed, we are ready to recover the file called `data_heterogeneity_original.csv` via `data_het_original_SAM_SUMmar26.R`. This script prepares the original heterogeneity dataset used for Figure 3 and the related sorting analysis. The key intuition is that it combines the firm-level treatment effects from the original LASSO predictions with a selected set of exporter characteristics, so that the treatment effects can later be ranked and related to observable firm attributes.

More specifically, the script loads the original `SUM` and `SAM` LASSO predictions, the corresponding bootstrap prediction files, the 2019 firm-level dataset, and the previously constructed `data_heterogeneity.RData`. It first merges the original `SUM` and `SAM` predictions by firm and month and computes the individual treatment effect as `pred_SAM - pred_SUM`. It then reorganizes and enriches the heterogeneity dataset by grouping some industry and regional categories, generating polynomial terms for several continuous variables, and removing variables that are redundant, overly sparse, or not intended for the final heterogeneity specification. After selecting the relevant predictors, it merges these firm characteristics with the original treatment-effect data. The script also constructs the regression formulas that will later be used in the heterogeneity analysis, although in the attached version it stops at exporting the final prepared dataset rather than running the full OLS step.

The main input files are

- `final_data_19.RData`, `SUM_preds_lasso.RData`
- `lasso_preds_19.RData`
- `lasso.pred_SUM_boot_subsamp_cvlam_marzo26.RData`
- `lasso.pred_SAM_boot_subsamp_cvlam_marzo26.RData`
- `data_heterogeneity.RData`

The main output file is `data_heterogeneity_original.csv`.

The corresponding bootstrap version is obtained via `data_het_boots_SAM_SUMmar26.R`, which generates `data_heterogeneity_boots.csv`. Specifically, `data_het_boots_SAM_SUMmar26.R` constructs the bootstrap heterogeneity dataset used for Figure 3 and the sorted-effects analysis. The key idea is to take the bootstrap prediction draws from the `SUM` and `SAM` LASSO models, align them by firm, month, and bootstrap replication, and then merge them with the selected exporter characteristics. This makes it possible to study not only the average treatment effect distribution, but also how that distribution changes once bootstrap uncertainty is taken into account. More specifically, the script first recreates the original firm-level treatment effects by merging the original `SUM` and `SAM` predictions and computing `TE_indiv = pred_SAM - pred_SUM`. It then prepares the heterogeneity covariates by aggregating some industry and regional categories, creating polynomial terms for several continuous variables, and removing variables that are redundant or not used in the final

heterogeneity specification. After this preparation step, it reorganizes the bootstrap prediction files so that each bootstrap replication forms one combined dataset with both **SUM** and **SAM** predictions. These bootstrap predictions are then merged with the cleaned firm characteristics to produce the final bootstrap heterogeneity dataset.

The main input files are

- `SUM_preds_lasso.RData`
- `lasso_preds_19.RData`
- `lasso.pred_SUM_boot_subsamp_cvlam_marzo26.RData`
- `lasso.pred_SAM_boot_subsamp_cvlam_marzo26.RData`
- `data_heterogeneity.RData`
- `final_data_19.RData`

generated in previous scripts (see above).

The main output file is `data_heterogeneity_boots.csv`.

The CADiff analysis then proceeds in two steps: first `original_coefficients_mar26.do`, and second `cadiff_advanced_mar26.do`.

`original_coefficients_mar26.do` is the preliminary step. It takes the datasets constructed in the previous R scripts and prepares the baseline classification-analysis results used for Table 4. In particular, it uses the original and bootstrap heterogeneity datasets already created for the sorted-effects analysis, namely `data_heterogeneity_original.csv` and `data_heterogeneity_boots.csv`. Its role is to estimate the coefficients and corresponding uncertainty measures needed for the CADiff exercise, that is, the comparison between the more affected and less affected firms based on the **SAM-SUM** treatment-effect estimates. The file therefore uses as inputs the outputs of the previous R scripts, and produces the intermediate coefficient/results objects that are then used by the second Stata file.

`cadiff_advanced_mar26.do` is the final step that turns those estimated coefficients into Table 4. Using the intermediate results produced by `original_coefficients_mar26.do`, it computes the estimated differences in means between the group of more affected firms and the group of less affected firms for the treatment effect and for the additional firm covariates shown in the table. It also applies the final specification choices across columns, such as the inclusion of sector controls and month controls, and formats the final CADiff output reported in Table 4.

The main inputs for this two-step Table 4 procedure are therefore the outputs generated in the previous sections:

- `data_heterogeneity_original.csv`
- `data_heterogeneity_boots.csv`

These two files are themselves constructed from previously generated objects, including:

- `SUM_preds_lasso.RData`
- `lasso_preds_19.RData`
- `lasso.pred_SUM_boot_subsamp_cvlam_marzo26.RData`
- `lasso.pred_SAM_boot_subsamp_cvlam_marzo26.RData`
- `data_heterogeneity.RData`
- `final_data_19.RData`

The difference across the three columns lies only in the set of controls included in these regressions:

- **Column (1): no sector or month controls.** `original_coefficients_mar26.do` computes the original differences in means without adding sector or month controls. It saves the corresponding point estimates in `original_coefficients_advcaffdiff_nocontrol.dta`. Then `caffdiff_advanced_mar26.do` repeats the same exercise bootstrap by bootstrap using `data_heterogeneity_boots.csv`, computes the bootstrap standard deviation of each coefficient, merges it with the original estimate, and exports the final Table 4 results for column (1).
- **Column (2): sector controls included.** The same procedure is repeated, but now each regression includes sector dummies as controls. The original point estimates are saved in `original_coefficients_advcaffdiff_control_sec.dta`. The bootstrap script then reproduces the same controlled regressions in each bootstrap replication, computes the corresponding standard errors, and exports the final results for column (2).
- **Column (3): sector and month controls included.** The same difference-in-means exercise is run again, but this time controlling for both sector dummies and month dummies. The original point estimates are saved in `original_coefficients_advcaffdiff_control_months_sec.dta`. The bootstrap script mirrors this specification across bootstrap replications, computes standard errors and p-values, and exports the final results for column (3).

The main inputs of this two-step Table 4 procedure are the outputs generated in the previous scripts:

- `data_heterogeneity_original.csv`
- `data_heterogeneity_boots.csv`

These files are themselves constructed from previously generated objects, including:

- `SUM_preds_lasso.RData`

- `lasso_preds_19.RData`
- `lasso.pred_SUM.boot.subsamp.cvlam.marzo26.RData`
- `lasso.pred_SAM.boot.subsamp.cvlam.marzo26.RData`
- `data_heterogeneity.RData`
- `final_data_19.RData`

IMPORTANT: The file `joint_p_values_github.pdf` explains into details the construction of the `joint_p_values`. The user should cite the paper also in case she uses only the latter as it constitutes a novelty per se.

Reproducing Fig.5 Figure 5 is produced from the two heterogeneity datasets already constructed for the previous analysis, namely `data_heterogeneity_original.csv` and the file `data_heterogeneity_boots.csv`. The script `fig_5.R` uses these files to compare, month by month, the average predicted effect under SAM-SUM with the average predicted effect under Y-SUM, both based on the Logit-LASSO predictions.

More specifically, the script first loads `data_heterogeneity_original.csv` and computes the monthly point estimates from the original sample. For each month, it calculates: (i) the average treatment effect SAM-SUM, given by `TE_indiv`; and (ii) the average difference Y-SUM, given by the realized export outcome minus the SUM prediction. It then loads `data_heterogeneity_boots.csv` and, for each bootstrap replication and month, recomputes the corresponding monthly averages of SAM-SUM and Y-SUM. These bootstrap replications are used to estimate the standard error of each monthly mean and to construct 95% confidence intervals. Finally, the script plots the two monthly series together with their confidence intervals.

By default, `fig_5.R` looks for these two input files in the sibling folder `Tab.4`. If the files are stored elsewhere, the input folder can be specified by setting the environment variable `OBES_TAB4_DIR`. The output folder can analogously be specified through `OBES_FIG5_OUT_DIR`.

The main input files are:

- `data_heterogeneity_original.csv`
- `data_heterogeneity_boots.csv`

These inputs are outputs of the previous scripts described above, in particular:

- `data_het_original_SAM_SUMmar26.R`, which produces `data_heterogeneity_original.csv`
- `data_het_boots_SAM_SUMmar26.R`, which produces `data_heterogeneity_boots.csv`

The main output is Figure 5 itself, showing for each month:

- the average SAM-SUM effect,
- the average Y-SUM effect,

- and the corresponding 95% bootstrap confidence intervals.

The script saves the figure as `intervals_lasso_month_Y_SUM_r3.pdf` and `intervals_lasso_month_Y_SUM_r3.png`. It also saves the monthly estimates and confidence intervals in `fig5.monthly_effects.csv`.

Reproducing Fig.6 Figure 6 is based on the original heterogeneity dataset, specifically `data_heterogeneity_original.dta`, which is the Stata version of the dataset previously constructed for Figures 3 and 5. The code uses this file to compare firms according to the distribution of the `SUM` prediction error and to plot, by quarter, the average values of `SAM-SUM` and `Y-SUM` across percentiles of that distribution.

More specifically, the script first loads `data_heterogeneity_original.dta` and defines two main quantities: `effect = pred_sam - pred_sum`, corresponding to `SAM-SUM`, and `effectXX = export_future - pred_sum`, corresponding to `Y-SUM`. It then constructs the `SUM` prediction error, ranks firms by the percentile of this error, and compares the bottom and top quartiles of the distribution. The code also reports descriptive statistics and t-tests for selected firm characteristics across these groups.

The figure itself is then constructed quarter by quarter. For each quarter, the script sorts firms by the percentile of the `SUM` prediction error within that quarter, computes the average `SAM-SUM` effect and the average `Y-SUM` effect for each percentile, and plots the two resulting percentile profiles. The paper reports the first two quarterly panels, `ppercentilesq1.png` and `ppercentilesq2.png`.

By default, `figure.6.do` looks for the input file in the sibling folder `Tab.4`. If the file is stored elsewhere, the input folder can be specified by setting the environment variable `OBES_TAB4_DIR`. The output folder can analogously be specified through `OBES_FIG6_OUT_DIR`. The script saves both the quarterly PNG files and the corresponding percentile-level values in CSV files named `ppercentilesq1_values.csv`, `ppercentilesq2_values.csv`.

The main input file is:

- `data_heterogeneity_original.dta`

This Stata dataset is the output of the previous heterogeneity-construction step, and corresponds to the Stata version of `data_heterogeneity_original.csv`, which is produced by `data_het_original_SAM_SUMmar26.R` from:

- `SUM_preds_lasso.RData`
- `lasso_preds_19.RData`
- `data_heterogeneity.RData`
- `final_data_19.RData`

The main outputs are the quarterly percentile plots listed above, together with the descriptive statistics and t-test comparisons reported in the Stata output.

Reproducing Fig 7:Y_minus_SUM. The latter code is localized in Fig 4 and 7 in the subfolder called Y_minus_SUM. Figure 7 follows the same sorted-effects workflow as Figure 4, but it only requires the shock-unaware SUM predictions and their bootstrap draws for the final calculation. In particular, the original shock-unaware predictions SUM_preds_lasso.RData are produced by SUM_predictions_19_20_SUM.R, while the corresponding bootstrap prediction draws lasso.pred_SUM_boot_subsamp_cvlam_marzo26.RData are generated by the R code bootstrap_wlambdacv_and_subsampling_mar26.R. These are the SUM inputs already described for Figure 4.

The difference is only in the definition of the treatment effect used in the final plotting step. While Figure 4 studies the monthly sorted partial effects of SAM-SUM, the script for Figure 7 defines the effect as Y-SUM, that is, the realized export outcome minus the shock-unaware prediction. Concretely, in the original sample and in each bootstrap replication, the script replaces the treatment effect with `TE_indiv = export_future - pred_SUM`, while keeping the same procedure for computing the monthly SPE and APE curves and the same bootstrap confidence bands. The final plotting routine is otherwise unchanged. The script implementing this step is SPE_Y_SUM_mar26.R

The main input files are therefore:

- SUM_preds_lasso.RData
- lasso.pred_SUM_boot_subsamp_cvlam_marzo26.RData

These files are produced in the same previous scripts already referenced for Figure 4:

- SUM_predictions_19_20_SUM.R
- bootstrap_wlambdacv_and_subsampling_mar26.R

By default, SPE_Y_SUM_mar26.R looks for these inputs in the sibling folder SAM_minus_SUM. If the files are stored elsewhere, the input folder can be specified by setting the environment variable OBES_FIG7_INPUT_DIR. The output folder can analogously be specified through OBES_FIG7_OUT_DIR. The main output is Figure 7 itself, namely fig_7_goodlegend.pdf, together with fig_7_y_sum.pdf, a PNG copy, and the underlying values in fig7_spe_values.csv.

Reproducing Table 5 Content of Tab_5 folder.

Table 5 follows the same CADiff procedure described for Table 4, but applied to the Y-SUM effect instead of the SAM-SUM effect. The key difference is therefore in the definition of the treatment effect: here the relevant variable is Y-SUM, computed as the realized export outcome minus the shock-unaware prediction, that is, `export_future - pred_SUM`, rather than `pred_SAM - pred_SUM`.

As in the Table 4 workflow, the first step is to construct the original and bootstrap heterogeneity datasets for this alternative effect. The script data_het_original_Y_SUMmar26.R creates the original dataset by combining the original SUM predictions with the firm-level heterogeneity variables and then defining the treatment effect as `TE_indiv = export_future - pred_SUM`. Its output is data_heterogeneity_original_y_sum.csv. The companion script

`data_het_boots_Y_SUMmar26.R` performs the analogous construction for the bootstrap replications, aligning the bootstrap prediction draws with the same exporter characteristics and producing `data_heterogeneity_boots_y_sum.csv`. The logic of the data preparation is otherwise the same as in the SAM-SUM case.

By default, these R scripts infer all paths from the location of the `Tab_5` folder. In particular, they look for the final dataset in `data/data_out`, the SUM and SAM prediction objects in `Fig_4_and_7/SAM_minus_SUM`, and the heterogeneity object in `Tab_4/construction_data_heter`. If the replication package is reorganized locally, these locations can be overridden by setting the environment variables `OBES_TAB5_DIR`, `OBES_REPLICATION_DIR`, `OBES_DATA_OUT_DIR`, `OBES_SAM_SUM_DIR`, and `OBES_TAB4_HETER_DIR`.

These two datasets are then used in the Stata CADiff procedure for Table 5, exactly as in Table 4. First, `original_coefficients_Y_SUM_mar26.do` computes the original differences in means between the more affected and less affected firms under the Y-SUM classification. Second, `cadiff_advanced_Y_SUM_mar26.do` reproduces the same exercise bootstrap by bootstrap in order to obtain standard errors, p-values, and significance markers for the final table.

The Stata files are also path-portable. They can be run from inside `Tab_5`, or the `Tab_5` path can be passed as the first argument to the `do` command. All intermediate files and final `.tex` outputs are then written back to that folder and to its `boots_Y_SUM_mar26` and `advregs_Y_SUM_mar26` subfolders.

The main inputs (of the `.do` files) are therefore:

- `data_heterogeneity_original_y_sum.csv`
- `data_heterogeneity_boots_y_sum.csv`

These files are themselves constructed from previously generated objects, namely:

- `SUM_preds_lasso.RData`
- `lasso_preds_19.RData`
- `lasso.pred_SUM_boot_subsamp_cvlam_marzo26.RData`
- `lasso.pred_SAM_boot_subsamp_cvlam_marzo26.RData`
- `data_heterogeneity.RData`
- `final_data_19.RData`

The difference across the three columns is the same as in Table 4 and concerns only the included controls:

- **Column (1): no sector or month controls.**
- **Column (2): sector controls included.**
- **Column (3): sector and month controls included.**

Thus, Table 5 is obtained by repeating the same two-step CADiff exercise as in Table 4, but replacing the SAM-SUM treatment effect with the Y-SUM treatment effect throughout.

Code in section_4_point_5 The folder `section_4_point_5` contains the month-specific causal machine learning code used for the single-split figures and diagnostic appendix figures in Section 4.5 of the paper. The main entry points are the scripts `blp_clan_gates_month1.R`, ..., `blp_clan_gates_month12.R`. Each of these files is a light wrapper around the shared runner `run_section_4_point_5_month.R`. For each month, the runner pools the 2018 and 2019 firm-level datasets, defines the treatment indicator distinguishing the two years, preprocesses the covariates, estimates CATEs using the S-learner, T-learner, R-learner, DR-learner, and Causal Forest, constructs AIPW pseudo-outcomes, and saves the outputs used for Figure 9 and Appendix Figures 1–3. It also saves the single-split learner-specific CATE objects in `CLANS/`; the repeated-split CLAN exercise reported in Table 8 is implemented separately in `Tab_8`.

The main input files are:

- `final_data_18.RData`
- `final_data_19.RData`

These files are the same final firm-level datasets constructed in the earlier data-preparation step.

By default, the scripts look for these datasets in `data/data_out/` relative to the replication-code root and save their outputs in `section_4_point_5/`. If the data are stored elsewhere, the user can set `OBES_DATA_OUT_DIR`; if the replication-code root is not the parent folder of `section_4_point_5`, the user can set `OBES_REPLICATION_DIR`. The overall output folder can be changed with `OBES_SECTION45_OUTPUT_DIR`. The GATES, AIPW, propensity-score, and CATE-object folders can also be changed separately with `OBES_GATES_OUTPUT_DIR`, `OBES_EFFECT_PLOTS_DIR`, `OBES_PROPNENSITY_PLOTS_DIR`, and `OBES_CLANS_DIR`. The monthly scripts are computationally intensive because they estimate several machine-learning learners month by month.

The relationship with the later outputs is the following. Figure 8 is produced by the dedicated script `Fig_8/fig_8_jan.R`. Figure 9 and Appendix Figure 1 are based on the GATES component of the monthly workflow in `section_4_point_5`. Appendix Figure 2 uses the DML-AIPW component of the same monthly workflow, and Appendix Figure 3 uses the estimated 5-fold Random-Forest propensity scores. Table 6 and Table 7 provide the corresponding repeated-split BLP and GATES summaries using their dedicated scripts in `Tab_6` and `Tab_7`. Table 8 has its own repeated-split CLAN/CADiff workflow in `Tab_8`.

Reproducing Figure 8 Figure 8 is reproduced by the file `Fig_8/fig_8_jan.R`. The script starts from `final_data_18.RData` and `final_data_19.RData`, pools the two years, defines the treatment indicator, and focuses on January. It randomly splits the observations into an auxiliary sample A and a main sample M , estimates the January T-learner in A , and predicts the treatment-effect distributions in M . The plotted distributions are the CATE, the CATT based on $\hat{\alpha}$, the CATT based on $\hat{\hat{\alpha}}$, and the CATU, matching the comparison described in Section 4.5 of the paper.

The main output is:

- `CATE_CATU_CATT.png`, saved by default in the folder `Fig_8/`.

As above, the default data location is `data/data_out/` relative to the replication-code root. The user can override it with `OBES_DATA_OUT_DIR`; the output folder for the figure can be changed with `OBES_FIG8_OUTPUT_DIR`.

Reproducing Table 6

Files contained in `Tab_6`.

Table 6 is reproduced by the file `tab_6.R`, which implements the Best Linear Predictor (BLP) validation exercise for the estimated CATEs across all months and learners. The script starts from the same two firm-level datasets used in the monthly heterogeneity analysis of `section_4_point_5`, namely `final_data_18.RData` and `final_data_19.RData`. It pools the two years, defines the treatment indicator `W` equal to one for 2019 and zero for 2018, harmonizes the common covariates, removes constant variables, converts selected factors and `iso` variables, and then builds a month-specific design matrix. For each month, the script repeatedly splits the sample into an auxiliary sample A and a main sample M . In each split, it estimates learner-specific CATEs in A using five methods—S-learner, T-learner, R-learner, DR-learner, and Causal Forest—and evaluates them on M . It then constructs the AIPW pseudo-outcome on the main sample and runs the BLP regression

$$\tilde{Y}^{ATE} = \beta_1 + \beta_2(\hat{\Delta}(X) - E[\hat{\Delta}(X)]) + u,$$

where $\hat{\Delta}(X)$ is the learner-specific predicted CATE. For each month and learner, the script stores the estimated β_2 , its p-value, and the regression R^2 , and finally reports the median of these statistics across repeated random A–M splits. The online version of `tab_6.R` is configured with the final-paper settings: 100 repeated A–M splits, 2,000 trees for the random-forest learners, 2,000 trees for the causal forest, GRF tuning enabled, and 5-fold cross-fitting for the AIPW pseudo-outcome. These settings are computationally intensive; users who only want a quick smoke test can temporarily reduce the speed-control parameters documented in the script. The main inputs are therefore `final_data_18.RData` and `final_data_19.RData`, already introduced in the previous section on `section_4_point_5`. The main outputs are:

- `table6_blp_results.RData`, containing the raw and summary BLP results;
- `table6_blp_results.csv`, containing the long-format summary table;
- `table6_blp_display.csv`, containing the wide display table formatted like Table 6.

Reproducing Table 7 Table 7 is reproduced by the file `Tab_7.R`, which implements the GATES validation exercise for the estimated CATEs across all months and learners. As for Table 6, the script starts from the pooled 2018–2019 firm-level datasets `final_data_18.RData` and `final_data_19.RData`, already introduced in the description of `section_4_point_5`. It harmonizes the common covariates, defines the treatment indicator `W` equal to one for 2019 and zero for 2018, removes constant variables, converts selected factors and `iso` variables, and builds a month-specific design matrix. For each month, the script repeatedly splits the sample into an auxiliary sample A and a main sample M . In each split, it estimates learner-specific CATEs in A using five methods—S-Learner, T-Learner, R-Learner, DR-Learner, and Generalized Random Forest—and evaluates them on M . It then computes the AIPW pseudo-outcome on the main sample and partitions the predicted CATE distribution into $K = 4$

groups. For each learner and group, it estimates the corresponding Group Average Treatment Effect (GATES), that is, the average pseudo-outcome within each predicted-CATE quartile. The script stores the group-specific coefficients and p-values for every split, and finally reports their median across repeated random A–M splits. The online version of `Tab_7.R` is configured with the final-paper settings: 100 repeated A–M splits, 2,000 trees for the random-forest learners, 2,000 trees for the causal forest, GRF tuning enabled, and 5-fold cross-fitting for the AIPW pseudo-outcome. These settings are computationally intensive; users who only want a quick smoke test can temporarily reduce the speed-control parameters documented in the script. The main inputs are therefore `final_data_18.RData` and `final_data_19.RData`. The main outputs are:

- `table7_gates_results.RData`, containing the raw and summary GATES results;
- `table7_gates_results.csv`, containing the long-format summary table;
- `table7_gates_display.csv`, containing the wide display table formatted like Table 7.

Reproducing Figure 9 Figure 9 is reproduced from the GATES outputs generated by `section_4_point_5`. Run the month-specific scripts for January, April, and October:

- `blp_clan_gates_month1.R`
- `blp_clan_gates_month4.R`
- `blp_clan_gates_month10.R`

Each script estimates the five CATE learners in an auxiliary sample, computes the AIPW pseudo-outcome in the main sample, divides the predicted CATE distribution into four groups, and plots the GATES estimates across methods. The three figure files used in the paper are:

- `gates_january_r4_NEW.png`
- `gates_april_r4_NEW.png`
- `gates_october_r4_NEW.png`

The scripts also save the corresponding `gates_<month>.r.pdf`, `gates_<month>.values.RData`, and `gates_<month>.values.csv` files.

Reproducing Figure Appx.1 Figure Appx.1 is the appendix version of Figure 9 and reports the same GATES plots for all months from January to December. To reproduce it, run all scripts `blp_clan_gates_month1.R`, ..., `blp_clan_gates_month12.R`. The main outputs are:

- monthly PDF files named `gates_<month>.r.pdf` for January through November
- `Gates_december_r.pdf`, matching the filename used in the manuscript
- `gates_<month>.values.csv` and `gates_<month>.values.RData`, containing the plotted coefficients and confidence intervals.

Reproducing Figure Appx.2 Figure Appx.2 is reproduced by the DML-AIPW part of the same monthly scripts, corresponding to the block labelled `Double ML for AIPW with causalDML package` in the original working code. For each month, the runner estimates the DML-AIPW ATE with 5-fold cross-fitting and nuisance parameters estimated with Random Forest, then computes the corresponding ATT using the same nuisance-parameter estimates. The scripts save one plot per month in:

- monthly PDF files in `effect_plots/`, named `effect_plot_month_<m>r.pdf` for $m = 1, \dots, 12$
- `effect_plots/effect_plot_month_<m>r.values.csv`, containing the plotted ATE and ATT estimates, standard errors, and confidence intervals.

Reproducing Figure Appx.3 Figure Appx.3 is reproduced by the propensity-score part of the monthly scripts, corresponding to the block labelled `Estimated propensity score with RF with 5 folds` in the original working code. For each month, the runner uses the 5-fold Random-Forest nuisance model from `causalDML` and plots the estimated propensity-score distributions separately for the 2018 control cohort and the 2019 treated cohort. The main outputs are:

- monthly PDF files in `propensity_scores/`, named `propensity_plot_month_<m>5_folds.pdf` for $m = 1, \dots, 12$
- `propensity_scores/propensity_plot_month_<m>5_folds.values.csv`, containing the plotted propensity scores
- `propensity_scores/propensity_plot_month_<m>5_folds.summary.csv`, containing group-specific summary statistics.

Reproducing Table 8 Table 8 is produced from the folder `Tab_8`, which is organized in two layers. The first layer is the folder `multiple_splits/`, containing one month-specific script for each month, namely `month1_multiple_splits.R`, ..., `month12_multiple_splits.R`. These scripts are the first step and must be run before the learner-specific CLAN scripts. As in the code described for `section_4_point_5`, each month-specific file loads `final_data_18.RData` and `final_data_19.RData`, harmonizes the common covariates across years, defines the treatment indicator `W` equal to one for 2019 and zero for 2018, preprocesses the month-specific sample, and repeatedly splits that monthly sample into an auxiliary sample A and a main sample M . The online scripts are configured with 100 repeated A – M splits, matching the final-paper setting. For each split, the script estimates CATEs using five learners—S-Learner, T-Learner, R-Learner, DR-Learner, and Causal Forest—and evaluates them on the held-out main sample. The resulting learner-specific predicted CATE datasets are saved in `multiple_splits/multiple_splits_datasets/` with filenames of the following form `cates_<learner>_<month>_learner_split<i>.RData`. The same scripts also save split-specific BLP and GATES objects, but for Table 8 the relevant intermediate outputs are the saved `cates_*` files.

The second layer consists of the learner-specific CLAN aggregation scripts, namely:

- `clans_multiple_splits_s_learner.R`,
- `clan_multiple_splits_R_learner.R`,
- `clan_multiple_splits_dr_learner.R`,
- `clan_multiple_splits_CF.R`.

Each of these scripts must be run after the month-specific scripts have generated the split-level `cates_*` files. A learner-specific script loads all split-level CATE datasets for that learner across months, appends them into a single dataset, reconstructs month dummies, aggregates detailed industries into the seven sector groups used in the paper, and defines the transport-mode indicators and other CLAN covariates. It then performs the classification analysis on the extreme quartiles of the learner-specific predicted CATE distribution, estimating the difference in means between the more affected and less affected firms under three specifications: without controls, controlling for sectors, and controlling for both sectors and months. For each covariate and specification, the script stores the coefficient and p-value split by split, and then reports their median across splits. The final outputs are saved in the folder `multiple_splits_results/` as learner-specific CSV files, including one file for each specification. For example, the Causal Forest script writes:

- `median_results_ordered_cf_learner_normal.csv`,
- `median_results_ordered_cf_learner_industry.csv`,
- `median_results_ordered_cf_learner_industry_month.csv`.

The analogous scripts for the S-, R-, and DR-learners write corresponding files with learner-specific names.

The main inputs of the full Table 8 procedure are therefore:

- `final_data_18.RData`
- `final_data_19.RData`

already introduced in the description of `section_4_point_5`. These are the same inputs used to construct the month-specific $A-M$ splits and learner-specific CATE predictions. The main intermediate outputs are the split-level files:

- `cates_s_<month>_learner_split<i>.RData`
- `cates_t_<month>_learner_split<i>.RData`
- `cates_r_<month>_learner_split<i>.RData`
- `cates_dr_<month>_learner_split<i>.RData`
- `cates_cf_<month>_learner_split<i>.RData`

saved in `multiple_splits/multiple_splits_datasets/`. The final outputs for Table 8 are the learner-specific median CLAN tables saved in `multiple_splits_results/`.

Hence, to reproduce Table 8, the correct order is: first run `month1_multiple_splits.R`, ..., `month12_multiple_splits.R` to generate the split-level learner-specific CATE files; second, run the learner-specific CLAN aggregation scripts to combine those files and compute the median CADiff/CLAN coefficients reported in the table.

5 Datasets Used

This section describes all datasets utilized in the analysis, including whether they are public or confidential and how they are used in the replication.

5.1 Data Availability and Access

All data sources used in this paper are summarized above, with details on their availability:

- **External Public Data (Not Included):** The core raw data used in this study consist of firm-level export and import transaction data from Colombian Customs, made available by the Colombian National Administrative Department of Statistics (DANE). These data are publicly accessible through the DANE microdata portal but are not redistributed as part of this replication package due to data usage and copyright restrictions.

Replicators can obtain the original data directly from the official source:

<https://microdatos.dane.gov.co/index.php/catalog/472/> (exports)

<https://microdatos.dane.gov.co/index.php/catalog/473/> (imports)

Rights and Permissions: The authors accessed the data through official public sources. However, they do not have permission to redistribute the original datasets. All data included in this replication package were generated by the authors and can be freely shared. Users are responsible for complying with the terms and conditions of the original data providers when accessing and using the raw data.

5.2 Public Data Sources

- *Colombian Customs Microdata (Exports and Imports):* Firm-level transaction data from Colombian Customs, provided by the Colombian National Administrative Department of Statistics (DANE). The data include export and import transactions by firms, with detailed information on products (HS codes), destinations, transport modes, and values.

Access: Publicly available at:

<https://microdatos.dane.gov.co/index.php/catalog/472/> (exports)

<https://microdatos.dane.gov.co/index.php/catalog/473/> (imports)

Usage: Used to construct the firm-level panel datasets and all main outcome variables in the paper.

6 Contact Information

For any questions regarding the replication files or if any issues arise in reproducing the results, please contact the author(s):

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7 Additional Notes

- **Randomness and Replicability:** The results involve simulations with random draws. For exact replication, the code sets a fixed random seed (e.g., `seed = 12345`) wherever applicable. Minor numerical differences might occur if a different software version is used.
- **Computational Intensity:** Some parts of the analysis are computationally intensive. For instance, the full estimation in Section 4 may take several hours. If running on a slower computer, expect longer runtimes or consider using a subset of data for testing.
- **Operating Environment:** The code has been tested in the environment described above. Different operating systems or software versions should generally work, but may require slight modifications (e.g., file path formats or memory settings).
- **Reproducibility:** All tables and figures in the paper should be exactly reproducible¹ with the provided code and data. If any result cannot be replicated due to external dependencies or data omissions, it is noted in the sections above.

¹With the caveat noted in Section 4

References

Hale, T., A. Petherick, T. Phillips, and S. Webster (2020). Variation in government responses to COVID-19. *Blavatnik school of government working paper 31*, 2020–11.